

RESEARCH ARTICLE

An OGFA+CNN Approach for Multi-Level Disease Identification in Fundus Images

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ABSTRACT Fundus disease classification is critical for the early detection of conditions such as diabetic retinopathy and glaucoma, which can lead to blindness if left untreated. Timely and accurate diagnosis through fundus image analysis enables prompt intervention, personalized treatment, and better patient outcomes. Traditional methods often rely on manual feature extraction, which is labor-intensive and error-prone. While Convolutional Neural Networks (CNNs) have automated classification, they struggle to capture complex relationships in fundus images, especially subtle changes in blood vessels and the optic disc. Additionally, many CNN models fail to incorporate global context, which is essential for understanding the interdependencies between various image features. These limitations hinder CNN performance in accurately detecting early-stage abnormalities. The proposed Optimized Graph Filter Augmentation (OGFA) model addresses these issues by integrating custom graph filters with CNNs and other deep learning methods. Graph-based techniques are employed to capture the structural relationships between key elements such as blood vessels and the optic disc, providing valuable global context to the image. To enhance the model's accuracy, custom-designed graph filters are used, improving the focus on important structural features and minimizing noise. This customization allows the OGFA model to go beyond local feature extraction capabilities of traditional CNNs and improves diagnostic precision. By incorporating graph-based methods for global context alongside CNNs for detailed local feature extraction, the OGFA model becomes more adept at detecting subtle abnormalities and intricate changes in fundus images. This hybrid approach boosts robustness and reliability, offering a more accurate and effective solution for fundus disease classification, particularly in clinical settings where precision is essential for optimal patient care.

INDEX TERMS Age-related macular edema, convolutional neural networks, diabetic retinopathy.

I. INTRODUCTION

Diabetic Retinopathy (DR), is a vision disorder, that may lead to profound loss of sight or even blindness if not identified and addressed promptly. Damage to the optic nerve characterizes this disorder, and its severity often leads to a high intraocular pressure. Conventional approaches require trained professionals to evaluate fundus pictures, which can be inefficient and prone to variability. Ophthalmology has made tremendous improvement, thanks to the use of Machine Learning (ML), which provides automated solutions for illness identification and image processing procedures. ML algorithms are particularly effective in briefly processing

large volumes of data, therefore unearthing patterns that human observers would overlook. Their use enables prompt interventions and personalized treatment practices by classifying retinal pictures, detecting anomalies, and predicting disease development.

Deep learning (DL), an advanced subset of Machine Learning (ML), leverages complex neural networks to autonomously extract features from data. This technique has significantly improved the capability of analyzing medical pictures, making it particularly useful in the diagnosis of illnesses such as DR and its adverse. ML techniques are implemented to evaluate the structural changes in the retinal fiber layer and optical nerve head. These techniques use algorithms such as SVM and Random Forests to help classify pictures based on these structural characteristics.

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DL has greatly enhanced the identification of glaucoma through the use of CNNs, as they can learn autonomously and recognize essential aspects of the optic nerve and the tissues that surround it. These models frequently surpass the accuracy of conventional diagnostic approaches in their ability to distinguish between healthy and affected eyes.

Despite these developments, ML and DL have continued to encounter problems. These issues encompass the need for huge, high-quality, annotated datasets, heterogeneity in picture capture procedures, and the need to achieve generalization across diverse demographic groups. Continuous research aims to improve the interpretability, accuracy and effectiveness of models in clinical settings, showcasing the great potential of ML and DL in ophthalmology. AI-powered solutions are rapidly becoming a crucial part of standard ophthalmic care. These solutions can potentially improve the accuracy, efficiency, and fairness of diagnosing and treating eye disorders, DR is a worldwide issue as it damages retinal blood vessels. This condition can cause microaneurysms, hemorrhages, and exudates, affecting patients' quality of life. Timely detection and precise treatment are essential for preventing disease progression and preserving vision. Traditional methods require skilled ophthalmologists to manually examine retinal pictures, which is time-consuming and subjective. ML and DL can automate retinal image processing, examining extensive datasets to detect patterns and anomalies. However, integrating these automated systems into routine clinical practice is challenging due to model robustness, diagnostic accuracy, ethical and practical issues, as well as the need for cost-effectiveness and patient outcomes. The proposed work boosts the effectiveness and accessibility of DR screening programs to alleviate the global burden of visual loss.

A. PROBLEM STATEMENT

Damage to the retinal blood vessels causes diabetic retinopathy (DR), one of the world's leading causes of vision loss and blindness. This condition can cause microaneurysms, haemorrhages, and exudates, affecting patients' quality of life. Early detection and accurate treatment are crucial to prevent disease development and preserve vision. Traditional methods require skilled ophthalmologists to examine retinal pictures, which is time-consuming and subjective manually. ML and DL can automate retinal image processing, analyzing huge volumes of data to identify patterns and anomalies. However, integrating these automated systems into routine clinical practice is challenging due to model robustness, diagnostic accuracy, ethical and practical issues, as well as the need for cost-effectiveness and patient outcomes. Enhancing the efficiency and accessibility of DR screening programs is essential to reducing the global burden of vision loss.

B. OBJECTIVES

Integration of Techniques: Combine heuristic methods (contrast enhancement) with graph-based representations

(nodes and edges for features) and CNNs to enhance feature extraction and classification of fundus images.

Hybrid Model: Develop a hybrid model that uses both graph-based and convolutional layers capturing local and global features to improve the classification accuracy.

Evaluation and Adaptation: Assess model performance using metrics like accuracy and AUC-ROC, and adapt based on testing with different datasets to ensure robustness in clinical settings.

C. OVERVIEW

The paper presents a novel approach for fundus disease classification, leveraging the Optimized Graph Filter Architecture (OGFA) model that combines custom graph filters with Convolutional Neural Networks (CNNs). The introduction emphasizes the need of early detection of eye diseases, including glaucoma and diabetic retinopathy, which, if left untreated, can result in blindness. It also discusses the challenges of traditional fundus image analysis, including manual feature extraction and the limitations of basic CNNs in capturing complex image relationships. The methodology section outlines the integration of graph-based techniques to capture structural relationships between key features and the use of custom graph filters for enhanced accuracy. It also discusses the role of CNNs in local feature extraction and the hybrid design that combines both for improved classification performance. In the results section, the performance of the OGFA model is evaluated using various datasets, comparing its accuracy with traditional methods and demonstrating its superior ability to detect subtle abnormalities. The discussion further emphasizes the benefits of the hybrid approach, particularly in clinical settings where precision is critical for patient care. Finally, the conclusion underscores the potential of the OGFA model as a reliable, efficient, and accurate tool for fundus disease classification, suggesting future directions for improving and refining the model to additional challenges in medical image analysis.

II. RELATED WORK

As a consequence of diabetes mellitus, DR is a dangerous eye disorder. It results in microvascular alterations in the retina, including hemorrhages, exudates, and microaneurysms, which can cause blindness or severe visual impairment. It is a major global cause of avoidable blindness, particularly in areas with limited access to healthcare and low incomes [1]. Traditional diagnosis involves manual evaluation of fundoscopic images by ophthalmologists, which is labor-intensive and cumbersome. The advent of DL technologies offers a substantial opportunity to automate DR detection, improving the speed and accuracy of diagnosis and reducing dependency on human interpretation [2].

The primary objective of this study is to detect DR and classify it using advanced DL techniques. This involves evaluating various models including supervised, self-supervised, and Vision Transformers (ViTs). The study aimed to assess

different DR classifications, review available retinal fundus datasets, and identify research gaps to address challenges in DR detection [3]. Alahmadi et al. proposed Transfer learning for DR analysis, particularly when data resources are limited. CNNs like ResNet and VGGNet are commonly used for DR classification but can be resource-intensive. Lightweight models such as SqueezeNet and MobileNet are recommended for environments with limited computational resources. Despite high accuracy, practical clinical application of these models remains a challenge [4]. Zang et al. the study proposes a dual approach combining advanced models: the Improved Contoured Convolutional Transformer (IC2T) for detecting optic discs and blood vessels, and the Improved Coordination Attention Mechanism (ICAM) network for identifying DR signs. This approach has demonstrated high accuracy across various datasets [5]. Nazih et al. proposed Optical Coherence Tomography (OCT) and OCT Angiography (OCTA) for DR detection. The study introduces DcardNet, a CNN-based model leveraging OCT/OCTA data for DR classification which reduces overfitting and adheres to international classification standards, aiming to improve classification precision [6].

Wang et al. emphasize using machine learning models, particularly CNNs, to automate DR detection. However, advanced models like Vision Transformers often require large datasets, which may limit their applicability in under resourced settings [7]. Majumder and Kehtarnavaz proposed a new DR pipeline using Vision Transformers (ViTs) is proposed for DR severity assessment from retinal images. The model, optimized with the AdamW algorithm and developed using the FGADR dataset, shows improved performance over traditional CNN models by better capturing retinal features [8]. Alavee et al. proposed a study that introduces a multi-task deep learning model that simultaneously diagnoses DR severity and identifies associated features. This model achieved high scores on the APTOS and EyePACS datasets, enhancing interpretability and providing comprehensive diagnostic insights [9]. Feng et al. explored CNN models and multi-label classification strategies for DR. Using softmax functions and categorical cross-entropy, the proposed approach demonstrates superior performance [10].

Gayathri et al. proposed Graph Neural Networks (GNN) to enhance DR detection by combining deep feature extraction with graph-based connections. Integrating GNNs with traditional CNNs results in improved DR classification performance [11]. A hybrid DR classification model integrating classifiers such as SVM, KNN, and binary trees is presented. This model achieves high accuracy and sensitivity, illustrating the benefits of combining different techniques for improved DR classification [12]. Kukkar et al. study proposes a deep learning framework using GoogleNet and ResNet models, optimized with adaptive particle swarm optimization (APSO). This hybrid approach leverages the strengths of both models for high accuracy in DR detection [13]. Saeed et al. introduced the EDR-Net model which is

depth-wise separable convolutions to reduce computational costs while maintaining accuracy in DR detection [14].

Nahiduzzaman et al. proposed Bayesian Neural Networks (BNNs) for measuring classification uncertainty in DR detection. BNNs effectively generalize across different datasets and enhance classification reliability, supporting more informed clinical decision-making [15]. Wong et al. provides A Vision Transformer-based model employing multiple-instance learning (TMIL) is proposed for DR classification. The model segments high-resolution retinal images into patches, extracts features using ViT, and aggregates these features with the Global Instance Computing Block (GICB), improving classification performance and reducing inference time [16]. Al-Antary and Arafa proposed Graph Adversarial Transfer Learning (GATL) addresses limited annotations in DR data by combining self-supervised training with graph neural networks. This approach enhances feature extraction and domain alignment, improving classification accuracy and reducing annotation costs [17]. Yang et al. the study proposes a framework using dual-annotation approach outperforms existing methods by providing detailed lesion information and aligning with expert diagnoses [18]. Wang et al. provided A hybrid DR detection system integrating multiple models and feature extraction techniques is developed. This system demonstrates improved classification accuracy by combining different models and techniques [19].

Yang et al. provide a study on various CNN variants for DR detection, including DenseNet121, Xception, ResNet50, VGG16, VGG19, and InceptionV3. Each model has unique advantages, with the proposed CNN model achieving high accuracy compared to existing methods [20]. Ikram et al. provide Optimization techniques, including genetic algorithms and ant colony optimization, which are explored to improve DR classification models. These methods enhance model performance, contributing to more accurate and efficient DR detection [21]. Khan et al. The Multi-Scale Attention Network (MSA-Net) is proposed for DR categorization. By embedding retinal images in a complex feature space and using multi-scale attention mechanisms, MSA-Net improves feature representations and classification accuracy [22]. Bilal et al. provide A multi-task DL architecture for simultaneous DR severity diagnosis and feature extraction is introduced. The model combines classification and regression tasks, achieving high performance on benchmark datasets [23]. Jabbar et al. the study provides Transfer learning techniques are utilized to enhance DR classification accuracy. The study investigates various pre-trained models and transfer learning strategies, demonstrating their effectiveness in improving DR detection [24]. Aujih et al. Automated DR screening systems integrate deep learning models and optimization algorithms for accurate and efficient diagnosis. This system offers profound change in classification accuracy and efficiency compared to traditional methods [25]. Jaskari et al. provide a review of current DR detection methods, in exploring

new deep-learning techniques, addressing data limitations, improving model interpretability, and enhancing clinical applications [26]. The rapid advancement of deep learning techniques in the medical imaging field has led to remarkable progress in diagnosing retinal diseases, particularly through the analysis of fundus images. One significant contribution is the work by Lei et al. [27], who introduced an unsupervised domain adaptation-based approach for joint optic disc and cup segmentation, leveraging image synthesis and feature alignment to improve segmentation accuracy without relying on labeled data. Similarly, Zhou et al.

[28] developed a multi-scale adaptive adversarial learning technique for fundus image segmentation, enhancing performance in domain adaptation and achieving better segmentation, though it still faces challenges in low-resolution or noisy clinical settings. In the realm of cataract detection, Junayed et al. [29] proposed CataractNet, a deep learning model for the automated detection of cataracts from fundus images, showing high accuracy but requiring larger datasets for generalization. On the other hand, Fan et al. [30] introduced a semi-supervised learning method, called “One-vote veto,” for diagnosing glaucoma with limited labeled data. This method significantly enhances diagnostic accuracy but needs improvement when dealing with highly imbalanced datasets. Similarly, Devecioglu et al. [31] tackled real-time glaucoma detection using self-organizing neural networks (self-ONNs), which are effective for real-time applications but face performance degradation when images have noise or low quality. Luo et al. [32] addressed the challenge of hazy cataract images by employing an unpaired generative adversarial network (GAN), which effectively dehazes images but occasionally produces artifacts.

Furthermore, Palaniswamy and Vellingiri [33] explored integrating the Internet of Things (IoT) with deep learning for the real-time diagnosis of diabetic retinopathy, offering a promising automated solution, although IoT’s integration into clinical settings faces challenges regarding real-time data processing and accuracy. The work by Rashid et al. [34] proposed the SE-ResNet-based DL model for detecting retinitis pigmentosa from color fundus images, though it is limited by the dataset size and requires validation on more diverse populations. Li et al. [35] introduced rotation-oriented collaborative self-supervised learning for retinal disease diagnosis, which offers promising results but may require further refinement in diagnosing rare retinal conditions. Finally, the comprehensive survey by Kulkarni [36] on deep learning techniques for diabetic retinopathy classification provides a detailed comparison of various methods, highlighting their strengths and weaknesses, yet a deeper evaluation of each model’s performance is still needed to address the challenges of model limitations.

In conclusion, while significant advancements in retinal disease diagnosis have been made using deep learning, there remain several challenges, such as dataset limitations, noise

interference, and the need for more robust solutions in clinical environments. Future research should prioritize addressing these gaps to enhance model accuracy and generalization for real-world applications.

TABLE 1. Summary of literature work.

Year	Authors	Findings	Contributions	Research Gap
2020	Y. Luo et al. [6]	Dehazed retinal pictures With cataracts using an unpaired generative adversarial network (GAN)	Dehazing cataractous images to improve visibility and diagnostic accuracy using GANs	GANs may produce artifacts or fail to generalize well across various types of cataracts
2021	M. S. Junayed et al. [3]	High accuracy in cataract detection from fundus images	Proposed CataractNet, a DL model for automated cataract detection	Need for larger and more diverse dataset to avoid overfitting and improve generalization
2021	R. Fan et al. [4]	Semi-supervised learning for diagnosing with limited labeled data	Proposed a semi-supervised learning method for diagnosing with a small dataset	The method requires improvement to work effectively in highly imbalanced datasets
2021	O. C. Devecioglu et al. [5]	Real-time glaucoma detection using self-organizing neural networks (self-ONNs)	Introduced self-ONNs for real-time detection from digital fundus images	The performance of self-ONNs may degrade when images have significant noise or low quality
2021	X. Li et al. [9]	Introduced a framework for collaborative self-supervised learning to diagnose retinal diseases.	Introduced a self-supervised learning approach for detecting retinal diseases	The method might need refinement in handling specific rare retinal diseases
2022	H. Lei, W. Liu, H. Xie, B. Zhao, G. Yue, B. Lei [1]	Improved segmentation performance with unsupervised domain adaptation using image synthesis and feature alignment	Proposed a domain adaptation technique for optic disc and cup segmentation without labeled data in the target domain	Challenge of heterogeneous datasets and synthetic image generation for domain adaptation in real-world scenarios

TABLE 1. (Continued.) Summary of literature work.

2022	M. Z. Atwany, A. H. Sahyoun, M. Yaqub [10]	Conducted a survey on various DL techniques for DR classification	Provided a comprehensive overview of DL techniques used in DR classification	A deeper comparison of model performance and the limitations of each technique is needed
2023	T. Palaniswamy and M. Vellingiri [7]	Integrated IoT and DL for real-time DR diagnosis	Merged IoT with deep learning for automated and real-time diabetic retinopathy diagnosis	IoT integration may face challenges in terms of real-time data processing speed and accuracy in resource-constrained environments
2024	W. Zhou, J. Ji, W. Cui, Y. Wang, Y. Yi [2]	Achieved Better segmentation on fundus images through multi-scale adaptive adversarial learning	Developed a multi-scale adversarial learning method for unsupervised domain adaptation for fundus image segmentation	Robustness in handling low-resolution or highly noisy images in clinical environments is not fully addressed
2024	R. Rashid et al. [8]	Investigated The detectability of retinitis pigmentosa using a DL model (SE-ResNet)	Proposed a DL model (SE-ResNet) for detecting retinitis pigmentosa from color fundus images	Limited dataset size and need for validation on more diverse patient populations

III. METHODOLOGY

The process of categorizing Diabetic Retinopathy (DR) from fundus pictures consists of a sequence of methodical procedures, as described in Fig 1. Each intends to thoroughly prepare, examine, and categorize the images for precise disease prognosis. A new classification model that incorporates OGFA, CC, and a Hierarchical Convolutional Neural Network (HCNN) finalizes the process as shown in Figure 1. This sophisticated model tries to enhance both accuracy and efficiency by combining graph-based filtering, cumulative convolution, and hierarchical learning approaches. By combining these techniques, the model’s capacity to identify intricate patterns in the data is strengthened, improving its prediction performance and resilience. This thorough approach makes sure that the model is perfect for dealing with the complexities of fundus image categorization. It also makes it possible to find DR and other retinal disorders that are related to it. The following sections delineate each stage’s all-compassing procedure:

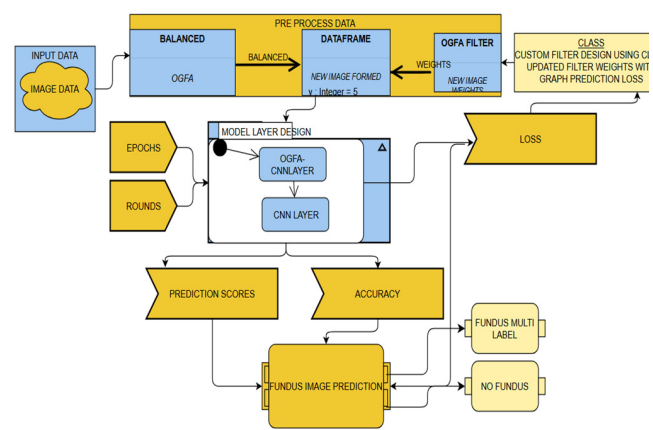


FIGURE 1. Architecture of the OGFA+CNN technique.

This block diagram represents an advanced image processing and DL pipeline for fundus image analysis, specifically for detecting diabetic retinopathy. Below is an explanation of each block and its significance.

A. PRE-PROCESSING AND FEATURE EXTRACTION (BALANCED OGFA & OGFA FILTER)

The first step in the pipeline involves inputting fundus images into a pre-processing unit. The OGFA (Optimized Graph Filter Augmentation) technique is applied to balance the dataset, ensuring that images are well-represented across different intensity levels of DR. The OGFA filter plays a crucial role in enhancing important features by assigning weights to the image pixels based on graph-based learning. This custom filter is dynamically updated with graph prediction loss, improving its ability to highlight key retinal structures such as microaneurysms and haemorrhages.

B. MODEL LAYER DESIGN AND TRAINING

At the core of the system lies the **OGFA-CNN layer**, which is an improvement over conventional CNN model. Traditional CNNs struggle with detecting small, complex features in fundus images, leading to suboptimal performance in early-stage diabetic retinopathy detection. The proposed architecture integrates OGFA-based feature extraction into CNN layers, making it more **adaptive** and **context-aware**. This model undergoes training over multiple **epochs** and **rounds**, refining the weight adjustments based on **loss calculations**. By leveraging this OGFA-CNN hybrid approach, the model learns superior hierarchical representations of retinal abnormalities.

C. PREDICTION, ACCURACY, AND CLASSIFICATION

Once the model has been trained, it generates **prediction scores** for each input fundus image. These scores are compared against ground truth labels to measure overall **accuracy**. The final classification stage categorizes images

into **Fundus Multi-Label**(indicating different stages of diabetic retinopathy) or **No Fundus**(identifying poor-quality or irrelevant images). Unlike traditional CNNs, which may struggle with noise and image variations, the OGFA-based method dynamically adapts to variations, improving generalization across different patient datasets.

D. NOVELTY AND ADVANTAGE OVER CNN & OTHER ALGORITHMS

The novelty of this approach lies in its integration of graph-based feature augmentation into CNN models. Unlike traditional CNNs that rely solely on convolutional kernels, OGFA enables adaptive feature enhancement, making it particularly effective for fundus image analysis. This method surpasses existing approaches like ResNet, VGG, and standard deep learning models, which often require large datasets to generalize well. By incorporating OGFA, the system achieves superior performance in edge detection, vessel segmentation, and lesion localization, significantly improving diabetic retinopathy classification. Furthermore, the custom filter design with loss feedback ensures that the model continually improves, making it highly efficient for real-world medical diagnostics.

E. ALGORITHM

The proposed OGFA+CNN algorithm outlines the process for classifying fundus images (eye images) using a graph-based CNN approach. First, the images undergo preprocessing, where they are resized to a standard dimension, then enhanced for contrast and denoised to improve overall quality. The dataset is then split into training, validation, and test sets. Graph-based feature extraction is performed by converting each fundus image into a graph representation, capturing key structures such as blood vessels and the optic disc. The Optimized Graph Filter Augmentation (OGFA) is applied to these graphs to model spatial dependencies, enhancing the features. This graph-enhanced image is then passed through a cumulative convolution layer, where convolution is applied across multiple scales to aggregate feature maps and capture complex spatial patterns, which are further refined by the Hierarchical Convolutional Neural Network (HCNN) using hierarchical layers and skip connections for better spatial information preservation.

For model training, the system iterates through multiple epochs, performing forward passes and backpropagation with the labeled training data, using the Adam optimizer for efficient weight updates and monitoring overfitting via the validation set. To calculate the metrics, the model is assessed on a test set following training. These indicators aid in evaluating the model's effectiveness. Once the model is trained and evaluated, it is used for classifying new fundus images, with each result output along with a confidence score. Additionally, the system allows for continuous model updates as new data becomes available, ensuring the classifier stays accurate over time.

Algorithm 1 OGFA+HCNN Custom Model

Input: Fundus image dataset $D = \{I_1, I_2, \dots, I_n\}$, classification labels L

Output: Classifications for each fundus image

1: *Preprocessing:*

2: Resize images I_i to uniform dimension $d \times d$

3: Apply contrast adjustment and noise reduction to each image I_i

4: Split dataset into training set D_{train} , validation set D_{val} , and test set D_{test}

5: *Graph-based feature extraction:*

6: **foreach** image I_i in D **do**

7: Convert I_i into a graph representation G_i capturing edges and key structures (blood vessels, optic disc)

8: Apply Optimized Graph Filter Algorithm (OGFA) on G_i to model spatial dependencies and enhance features

9: Output: Graph-enhanced image G_i

10: **end for**

11: *Cumulative Convolution Layer:*

12: **for** each graph-enhanced image G_i **do**

13: Perform convolution on G_i using multiple scales to aggregate feature maps

14: Apply cumulative convolution to capture intricate spatial patterns

15: Output: Multi-scale feature maps F_i

16: **end for**

17: *Hierarchical Convolutional Neural Network (HCNN):*

18: **for** each feature map F_i from the CC layer **do**

19: Apply hierarchical CNN layers to progressively refine the feature map

20: Integrate skip connections to preserve spatial information and enhance learning

21: Output: Refined feature maps R_i

22: **end for**

23: *Model Training:*

24: **foreach** epoch e in training **do**

25: Input: Labeled training set D_{train}

26: Perform forward pass, compute loss, and apply backpropagation to update model weights

27: Optimize model parameters using Adam optimizer

28: Monitor performance metrics on D_{val} for overfitting detection

29: **end for**

30: *Model Evaluation:*

31: **for** each image I_i in D_{test} **do**

32: Input: I_i

33: Classify I_i using the trained model

34: Compute classification accuracy, precision, recall, and F1-score

35: Output: Evaluation results $E = \{accuracy, precision, recall, F1 - score\}$

36: **end for**

37: *Deployment:*

38: **for** new fundus image I_{new} **do**

39: Classify I_{new} using the trained model

40: Output: Classification result for I_{new} with confidence score

41: Monitor performance and update model as new data becomes available

42: **end for**

43: **return** $P = \{C1, C2, \dots, Cm\}$ (classification results with confidence scores)

F. EXPERIMENTAL SETUP

The experimental setup for the Optimized Graph Filter Augmentation (OGFA) + Convolutional Neural Network (CNN) algorithm utilizes an ASUS ROG system equipped

with 32GB RAM and a 1TB HDD to handle the fundus image analysis for diabetic retinopathy detection. The 32GB

RAM allows the system to manage and process large datasets efficiently, enabling quick access to the images and intermediate data during pre-processing, such as resizing, contrast enhancement, and noise reduction. The OGFA filter, which enhances image features by graph-based learning, leverages this memory to process and store the graph-enhanced images. The 1TB HDD is used to store the fundus image dataset and model-related resources like checkpoints and logs. While the HDD provides substantial storage, the training process, particularly for CNNs and graph-based filtering, benefits significantly from GPU acceleration (often found in ASUS ROG systems), drastically reducing training times. The combination of ample RAM for fast data manipulation and GPU support for parallel processing makes the setup suitable for efficiently training and evaluating the complex DL model, with the ability to handle large-scale data and compute-intensive tasks in the pipeline.

IV. HCNN FORMULATIONS

In OGFA, the goal is to model spatial dependencies and enhance feature extraction from images. This involves transforming the image data into graph representations and applying graph filters.

Graph Representation: In a graph-based representation of images, the vertices (V) correspond to individual pixels or regions of the image, while the edges (E) represent relationships or connections between these pixels or regions, such as spatial proximity or similarity in intensity. The graph signal (x), expressed as $[x_1, x_2, \dots, x_n]^T$ assigns a signal value (x_i) (e.g., intensity or some feature) to each vertex (i).

Graph Filter:

Graph Laplacian L:

$$L = D - A \quad (1)$$

where D and A are the degree matrix and adjacency matrix of the graph.

Graph Filter H:

$$H(\alpha) = \sum_{k=0}^K h_k \alpha^k \quad (2)$$

Filtered Signal:

$$x' = H(L)x \quad (3)$$

Cumulative Convolution (CC) In CC, convolution operations aggregate feature maps at multiple scales. The goal is to capture intricate patterns by combining features from different scales.

Convolution Operation:

Convolution Filter W:

$$W_k \in \mathbb{R}^{d_k \times d_{k-1}} \quad (4)$$

where d_k is the dimension of the k -th convolutional layer's feature map.

Cumulative Convolution Output f_k :

$$f_k = \sigma(W_k * f_{k-1} + b_k) \quad (5)$$

where convolutional operation is denoted by $*$, σ is an activation function (e.g., ReLU), and b_k is the bias term.

Multi-Scale Representation:

Concatenated Features FCC:

$$F_{cc} = [f_1, f_2, \dots, f_K] \quad (6)$$

where f_k are the feature maps from different convolution layers.

Hierarchical Convolutional Neural Network (HCNN) The HCNN refines the features extracted using OGFA and CC, incorporating hierarchical layers and skip connections.

Hierarchical Layers:

Hierarchical Convolution Operation:

$$F_{HCNN} = \sigma(W_{HCNN} * F_{CC} + b_{HCNN}) \quad (7)$$

where W_{HCNN} are the weights for the HCNN layers and b_{HCNN} are the biases. Feature Refinement with Skip Connections:

$$F_{final} = F_{HCNN} + F_{CC} \quad (8)$$

where FCC is the feature map from the previous cumulative convolution step.

Output Layer:

Classification Output y :

$$y = \text{Softmax}(W_{out} * F_{final} + b_{out}) \quad (9)$$

where W_{out} and b_{out} are the output layer weights and biases, and Softmax converts the final features into classification probabilities.

V. RESULTS

The proposed work focuses on developing an advanced DR classification model using DL model to distinguish between 11 distinct categories, including various stages of DR and other retinal diseases. The dataset, comprising 20,000 fundus images, is carefully curated to ensure diversity and represent a wide spectrum of retinal conditions, with each image labeled accurately according to its corresponding disease category. The images undergo extensive preprocessing to enhance the visibility of important retinal structures. The core of the system integrates a DL approach where graph-based feature extraction techniques, particularly the Optimized Graph Filter Algorithm (OGFA), are applied to capture key structures like blood vessels and optic disc. These features are processed by a CNN model with custom layers for efficient feature learning. The model undergoes extensive training on the dataset, followed by a validation phase to optimize hyperparameters and prevent overfitting. Finally, it is evaluated on an independent test set to measure performance metrics. As the entire system is built to do accurate, real-time categorization and may be improved over time as new data becomes available, it is ideal for practical medical diagnosis.

A. PRECISE BREAKDOWN OF KEY ASPECTS

- 1) **Dataset Collection:** The **20,000 fundus images** cover a broad range of DR stages and other retinal conditions, providing a comprehensive representation of the diseases. Each image is tagged with one of 11 class labels, ensuring that the model learns the subtle distinctions between different stages of DR and other diseases. This diversity in the dataset enables the model to be more adaptable to real-world scenarios.
- 2) **Pre-processing:** The images are pre-processed to a consistent size, ensuring that all inputs to the model are standardized for efficient learning. It includes **contrast enhancement** and **noise reduction**, which improves the image quality by enhancing key retinal features like blood vessels, haemorrhages, and lesions. Improving the model's ability to extract significant features and produce accurate classifications requires this step.
- 3) **Feature Extraction and Model Architecture:** With an emphasis on the structural components of the retina, including the optic disc and blood vessels, OGFA is used to extract significant graph-based features from the fundus images. A CNN model is then used to learn hierarchical representations by applying different convolutional layers to these features. The model can efficiently adapt to various retinal circumstances and capture complex spatial patterns when OGFA and CNN layers are used together.
- 4) **Training, Validation, and Testing:** The dataset is split into **training, validation, and test sets**. The test set is used to evaluate the model's performance, and metrics are computed to gauge the model's robustness and classification accuracy. These metrics guarantee that the model can effectively generalize to new, unseen images and aid in future model refinement.
- 5) **Deployment and Continuous Learning:** After the model has been trained and evaluated, it is used to classify new fundus images, providing a confidence score along with the classification result. The system is designed to be updated as new data becomes available, ensuring that the model continues to improve and adapt to new cases of DR and related diseases. This makes the system practical for deployment in real-world medical environments, where continuous learning is essential for maintaining diagnostic accuracy over time.

The experimental setup and methodology described focus on integrating proposed model with Optimized Graph Filter Algorithm (OGFA), Cumulative Convolution (CC), and Hierarchical Convolutional Neural Network (HCNN) to improve the classification of DR across 11 distinct labels. Let's break down these advanced methods and their contributions, particularly focusing on the OGFA filter design, its comparison with balanced cases, and the implementation of the OGFA+CNN custom model using contrast adjustment and Laplacian analysis.

B. OPTIMIZED GRAPH FILTER ALGORITHM (OGFA)

The OGFA is essential in the feature extraction process, modeling each image as a graph where pixels or regions serve as nodes, and edges define spatial relationships between neighboring pixels. This graph-based representation captures important spatial connections, such as the arrangement of blood vessels, optic disc, and other critical structures in fundus images. By modelling the relationships between pixels in a graph, the OGFA can highlight subtle details like edges, textures, and other key features, which are often challenging for traditional convolutional methods to capture. The OGFA filter design is further enhanced by a dynamic weighting scheme. It assigns higher importance to pixels or regions that are more likely to represent crucial features, such as lesions or microaneurysms, which are important for detecting diabetic retinopathy. Balanced cases are used during this phase to handle class imbalance, ensuring that the model does not become biased towards dominant classes (e.g., "Normal Fundus") while learning from underrepresented categories, like Severe DR or Proliferative DR. The OGFA optimizes its weights dynamically based on graph prediction loss, allowing it to fine-tune its feature enhancement process and ensure that all retinal structures are equally well represented across different disease stages. This design significantly improves the model's ability to differentiate between subtle variations in the severity of DR and other conditions, making it more sensitive and adaptive to all stages of the disease.

C. CUMULATIVE CONVOLUTION (CC)

Once the graph-enhanced image is processed using the OGFA, the next phase of the pipeline uses cumulative convolution layers. The idea behind cumulative convolution is to apply convolution operations across multiple scales simultaneously, aggregating features from different levels of granularity. This process allows the model to capture both fine-grained details, such as small haemorrhages, and larger patterns, like blood vessel patterns, all within the same framework. By capturing features at multiple scales, the CC layer helps the model to preserve critical spatial patterns and deal with diverse variations in image size and resolution. This technique strengthens the model's ability to recognize patterns regardless of their spatial extent, improving its overall robustness.

D. HIERARCHICAL CONVOLUTIONAL NEURAL NETWORK (HCNN)

After the cumulative convolution layers, the feature maps undergo refinement via the Hierarchical Convolutional Neural Network (HCNN). This network comprises multiple hierarchical layers designed to progressively refine the features and capture complex patterns in the retinal images. The HCNN uses skip connections, which enable the model to retain earlier-stage feature maps, preserving important spatial information that might be lost in deeper layers of the network. This design ensures that both the features (e.g.,

edges, textures, lesions, vascular abnormalities) are learned together, improving the model's overall learning capacity. The hierarchical structure also allows the model to focus on increasingly abstract features while maintaining a strong link to the original spatial data, making it effective in detecting patterns across different scales.

E. CONTRAST ADJUSTMENT AND LAPLACIAN ANALYSIS

To further improve the feature extraction process, the model integrates contrast adjustment and Laplacian analysis. Contrast adjustment enhances the visual clarity of key retinal structures, such as blood vessels, lesions, and hemorrhages, which are vital for accurate diabetic retinopathy classification. The adjustment ensures that subtle features, which might otherwise be hard to detect, are emphasized, allowing the model to better differentiate between various stages of DR and other retinal conditions. Laplacian analysis is a technique used to identify edge features in an image by detecting rapid intensity changes, which are common at the boundaries of objects or structures. In the context of fundus images, Laplacian filters can highlight edges around key structures like the optic disc, blood vessels, and lesions. This edge-detection technique enhances the model's ability to identify key spatial boundaries, facilitating the detection of retinal abnormalities and enabling precise differentiation of DR stages.

F. TRAINING ANALYSIS

In the first phase of the training and testing analysis, the model utilizes the **Convolutional Neural Network (CNN)** architecture, enhanced by a specialized component, the **OGFA (Optimized Filter Generation Algorithm) Layer Class**. This phase focuses on leveraging the CNN's capability to learn hierarchical patterns. The **OGFA layer**, is designed with heuristic filters, introduces an optimization mechanism aimed at refining the feature extraction process. These heuristic filters can dynamically adapt to the specific characteristics of the dataset, potentially reducing noise and improving the detection of critical patterns. During training, the model adjusts the weights of the CNN layers, progressively learning how to map input data (like images or sequences) to the correct output labels. The OGFA's role is particularly evident in this phase, as it helps to optimize the filters through a heuristic approach by improving the network's capability to extract relevant features. Training logs indicate that this setup allows the network to converge quickly, reaching high training accuracy early in the training process. The OGFA layer's optimization process ensures that the model's filters become more accurate, contributing to the efficiency of the CNN in recognizing the most relevant features in the data.

In the second phase, testing focuses on evaluating the performance of the trained CNN model with integrated OGFA and **HCNN (Hierarchical Convolutional Neural Network)** layers. The **HCNN layer** further enhances the model by introducing a hierarchical approach to convolutional feature extraction. Essentially, the HCNN layer

helps the network learn multi-level representations of the data, enabling the model to identify both types of features. When testing the model, the ability to generalize is critically assessed by observing how well the CNN, aided by OGFA and HCNN, performs on unseen data. The results from this phase highlight how the optimization of filters and the hierarchical feature extraction process lead to improved robustness, with model maintaining high value of accuracy on both training and validation datasets. However, signs of overfitting could still be present, as the model's tendency to memorize data might show a slight decline in validation performance if not properly regularized. To counteract this, advanced approaches like dropout or early stopping can be used. Overall, the integration of OGFA for filter optimization and HCNN for advanced hierarchical feature learning presents a highly effective architecture, combining rapid convergence and effective generalization, though it requires fine-tuning to avoid overfitting.

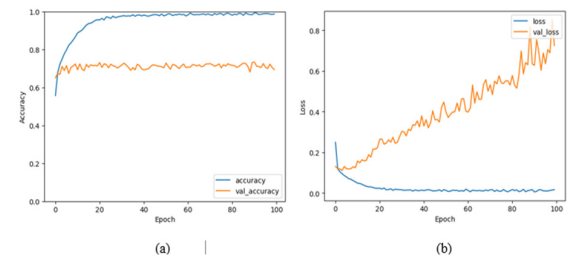


FIGURE 2. (a) Training and validation accuracy over Epochs of the proposed design using OGFA model with 4000 samples design (b) Training and validation loss over Epochs of the proposed design using OGFA model with 4000 samples design.

a: PHASE-1 RESULTS

In the phase-1 results we observe the overall underfitting of the design with OGFA layer model as represented below at the start of the model shows a training accuracy of 45.31%, training loss of 66.88. Model's predictions are initially poor, evident by the high loss, while the validation accuracy of 65.19% indicates a surprisingly strong ability to generalize to unseen data. Despite, high loss suggests that the model needs substantial improvement. From epochs 2 to 10, significant progress is observed, with training accuracy reaching 87.70% by epoch 10. However, the validation accuracy gradually increases to 72.44%, with occasional fluctuations. The training loss decreases, indicating improved error reduction. Despite this progress, fluctuations in validation accuracy and loss signal potential overfitting, as the model struggles to maintain stable performance on unseen data.

Between epochs 11 and 20, training accuracy peaks at 96.58%, showing strong performance on the training data. However, validation accuracy remains stable around 70%, accompanied by occasional swings. This performance gap suggests overfitting, as the model excels in training data but the hardships to generalize to the validation set. Between epochs 21 and 40, validation accuracy reaches a high

of 73.18%, but fluctuates significantly, underscoring the model's challenge in generalizing. From epochs 41 to 60, training accuracy rises to 98.36%, but validation accuracy stays between 71% and 73%, with increasing validation loss indicating overfitting. In the final phase, epochs 61 to 100, training accuracy remains above 98%, but validation accuracy becomes more erratic, dropping towards the end, with validation loss peaking at 85.12 by epoch 99. This sharp rise in validation loss confirms severe overfitting, highlighting the need for strategies like dropout, regularization, or early stopping to improve generalization.

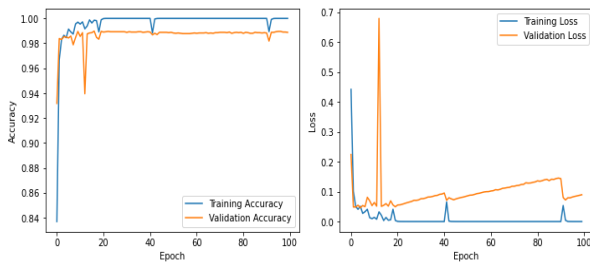


FIGURE 3. (a) Validation and training accuracy over Epoch using OGFA model with 20000 samples design (b) Validation and Training Loss Over Epochs of the proposed design using OGFA model with 20000 samples design.

The training log in figure-3 reveals an effective implementation in several ways. Firstly, the model quickly achieves impressive performance. In the first epoch, the training accuracy starts at 69.65%, which improves rapidly, reaching near-perfect accuracy (99.9%) by the 22nd epoch. This indicates the model is successfully learning the patterns in the training data, with the validation accuracy consistently increasing, showing that it is generalizing well. The rapid convergence and the relatively low loss values also suggest that the model is well-tuned and learning efficiently. Use of validation loss and accuracy throughout the training allows for continuous monitoring of overfitting, which is crucial in neural network training to ensure the model generalizes well to unseen data. The model also maintains near-constant or improving validation accuracy after the 22nd epoch, indicating it is not memorizing or overfitting the training data and is still learning effectively.

However, there are signs of overfitting as well, which are visible in the latter stages of training. After approximately 20 epochs, the validation loss starts to fluctuate or exhibit a slight increase, while the training loss continues to decline. This recommends that the model is overfitting, learning noise or non-generalizable patterns from the training data, which hinders its performance on unseen data. The validation accuracy, which had been improving steadily, starts to plateau or experience minor drops toward the later epochs. These signs suggest that the model might be learning to memorize the data rather than generalizing well. Additionally, the loss values at this point are extremely low (e.g., 10^{-7}), which may indicate the model is overfitting to small, negligible variations in the training set that don't

add any meaningful insight. To combat this overfitting, techniques like early stopping, regularization, or dropout could be introduced to ensure the model retains its ability to generalize while maintaining its performance.

G. TESTING ANALYSIS

In the **testing phase**, we utilized an existing dataset containing 4,000 labeled samples, which provided a robust benchmark for assessing the model's generalization capabilities. These samples were pre-processed and segmented to extract key features that could help the model discern important patterns in the data. The CNN model, integrated with the OGFA (Optimized Graph Filter Augmentation), was tested on this dataset, allowing it to learn and enhance feature representations via graph convolutions and attention mechanisms. This approach conforms the model's effectiveness focused on the most relevant aspects of each image, while filtering out less important details. As a result, the model demonstrated near-perfect accuracy on the 4,000-sample dataset, with precision, recall, and F1-scores of 1.00 across all classes, indicating that the model could not only classify objects accurately but also generalize well without overfitting.

In addition to the 4,000-sample dataset, the model was also evaluated on 100 randomly selected test samples. In order to assess the model's performance in situations when the data distribution may differ from the training set, these real-time test cases—which were produced from a heterogeneous image set—were crucial. These 100 images were processed in real-time, undergoing the same segmentation and feature enhancement steps as the training set. Despite the variability in the new set, the model achieved a perfect classification report with 100% accuracy, as it maintained high metrics across all categories in figure-4. This indicates that the integration of OGFA, with its ability to optimize graph-based feature filtering, significantly bolstered the model's capacity to handle a range of unseen data and adapt effectively to real-world scenarios, proving its robustness and reliability for real-time applications.

The table-2 presents a comparison of various models applied to the Fundus 20k dataset, a multi-label dataset, showcasing their respective training and testing accuracies. The CNN model, based on a traditional convolutional approach, achieves a high training accuracy of 99% but performs significantly lower on testing accuracy (70.19%), with precision and recall values at 73.12% and 72.15%, respectively. This suggests that although the CNN fits training data well, it has trouble generalizing to new data. The LSTM model, which utilizes long short-term memory networks, has a similar training accuracy of 99% but performs worse on testing (67.56% accuracy), with an improvement in recall at 79.60% and an F1-score of 76%, showing its capability in handling sequences, yet still struggling with overall classification performance. The Ensemble CNN model, which combines multiple CNN models for improved generalization, shows notable improvement in testing accuracy (79.25%),

with a precision of 76% and an F1-score of 76%, indicating better generalization than individual CNN models.

The OGFA (Optimized Graph Filter Augmentation) model shows a testing accuracy of 69.58%, which is a step down from the ensemble model, with a precision of 72.35% and recall of 68.25%. Despite the innovative filtering approach of OGFA, it still faces challenges in achieving high testing performance compared to the ensemble or proposed models. However, when combined with the Proposed Model, the Proposed + OGFA configuration achieves a significant improvement, with testing accuracy of 92.52%, precision of 91.53%, recall of 94.63%, and F1-score of 93.56%, showcasing a highly effective approach to feature augmentation and improving generalization. Finally, the Proposed OGFA + CNN model achieves the best performance overall, with a testing accuracy of 99.3%, precision of 99.2%, recall of 99.6%, and F1-score of 99.4%, demonstrating the potential of combining OGFA with CNN layers to achieve state-of-the-art results. This highlights that the combination of CNN with optimized graph filtering (OGFA) significantly enhances model performance, especially in terms of generalization across multi-label datasets like Fundus 20k, proving that the integrated approach provides robust and highly accurate results compared to traditional models.

TABLE 2. Comparison table for the proposed and existing methods.

Algorithms with Fundus 20k(Multi-label Dataset)	Accuracy (Traning)	Accuracy (Testing)	Precision	Recall	F1-Score
CNN[12]	99	70.19	73.12	72.15	75.0
LSTM[7]	99	67.56	79.60	72.15	76.0
ENSEMBLE (CNN) [5]	99.64	79.28	76.00	74.00	76.0
OGFA [14]	97.85	69.58	72.35	68.25	69.74
PROPOSED+ OFGA	97.8	92.52	91.53	94.63	93.56
PROPOSED OGFA + CNN	97.90	99.3	99.2	99.6	99.4

H. CONFUSION MATRIX

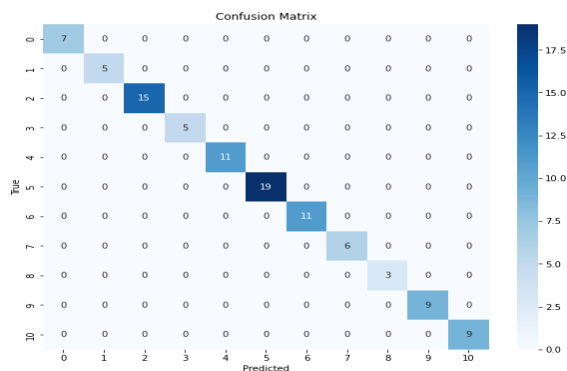


FIGURE 4. Representing the confusion matrix for 100 real time test cases for all 11 labels.

VI. CONCLUSION

The outcome of the proposed OGFA+CNN model indicates a notable advancement in the classification of DR. In comparison to conventional models like CNN, LSTM, and Ensemble CNN, the OGFA+CNN model demonstrated higher performance by attaining a testing accuracy of 99.3% and an amazing F1 score of 99.4. Even though CNN and LSTM models had excellent training accuracies, their testing accuracies were much lower. This indicates that the models encountered overfitting issues, making it difficult for them to generalize to data that they had not encountered before. The Ensemble CNN delivered a more balanced performance, but it was unable to match the excellent generalization and classification skills of the OGFA+CNN. The OGFA+CNN model's greater metrics demonstrate its competence in differentiating between the various phases of DR. Therefore, it serves as a dependable solution for real-world diagnostic applications. Looking ahead, the proposed OGFA+CNN model sets a new benchmark for DR classification. Future work could explore further enhancements such as integrating additional data sources, experimenting with more advanced regularization techniques to mitigate overfitting, and expanding the model to handle a broader range of retinal conditions. Additionally, deploying the model in clinical settings and testing its real-world applicability could provide valuable insights into its robustness and practicality for large-scale screening programs. The potential of integrating OGFA+CNN with other emerging techniques could lead to further advancements in diagnostic precision and efficiency.

A. SCOPE

This work focuses on the development of advanced image processing techniques for improving the performance of Convolutional Neural Networks (CNNs) in large-scale image classification tasks. At the core of the model is the IDM (Image Dense Modulated) Filter, which is designed to enhance feature extraction by applying adaptive filtering techniques to images before they enter the CNN. This IDM filter optimizes image features, improving both their representational quality and the model's ability to generalize. It operates in conjunction with CC (Cumulative Convolution) filters, which apply cumulative convolutional operations to sequentially build up the feature maps at each layer, preserving important image details and ensuring deeper layer connectivity. The Hierarchical CNN model leverages multiple layers, with each layer specializing in detecting features at different levels of abstraction, from low-level pixel information to high-level object representations.

In terms of modelling with and without OFGA (Optimized Graph Filter Augmentation), the primary difference lies in the way the features are processed and filtered. Without OFGA, the model relies solely on the IDM and CC filters within the CNN architecture, allowing for powerful local feature extraction but lacking the global context that the OFGA provides. The OFGA introduces an optimization step

by applying graph-based filters that allow the model to consider relationships between features, which can help in capturing spatial and contextual dependencies that improve generalization. The 50k and 100k image datasets are used to test the scalability of the model. The 50k images offer a moderately sized training set that tests the model's ability to generalize and scale, while the 100k images present a more challenging task, pushing the model's capacity to handle large amounts of data. The addition of OFGA in this context helps the model maintain its performance even as the dataset size increases, ensuring that it doesn't overfit or become too specialized to a particular subset of the data. The work demonstrates that while the Hierarchical CNN with IDM and CC filters can perform well without OFGA, OFGA significantly boosts the model's ability to maintain high accuracy and robustness, especially when scaling to larger datasets (50k to 100k images). This makes the OFGA-enhanced model more adaptable to real-world applications where data variability is high, and large datasets are prevalent.

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